Grant Selection Design and Start-Up Innovation: Evidence from Design-Dependent Subsidy Effects

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Curriculum Vitae

- **School**
  Sidcot Independent School England 2012–2014
  International Baccalaureate
- **University**
  Technical University of Munich B.Sc. Technology and Management 2014–2020
  Technical University of Munich M.Sc. Technology and Management 2020–2024
- **Professional Experience**
  CoInvest Corporate Finance Analyst 2021–2024
  agentix GmbH Co-Founder & COO 2024–2025
  Independent Management Consultant 2024–present

Master’s thesis

Standardising convertible loan agreements for start-ups

- analysed convertible loans as a financing instrument for young firms
- compared legal, financial, and contractual design choices
- assessed whether standardisation can reduce complexity and transaction costs
- developed a model convertible loan agreement

Link to the PhD proposal: how institutional design shapes financing, incentives, and firm behaviour.

Why this PhD topic fits my profile

My background connects **firm finance**, **innovation**, and **public funding**.

- **Master's thesis:** empirical work on firm finance and investment
- **agentix GmbH:** Co-Founder and COO of a Munich-based AI automation start-up
- **EXIST grant:** direct exposure to public start-up funding

Do the signals used to select firms generate the value public funding is meant to create?

- Which observable signals enter the funding decision?
- Do these signals predict innovation, spillovers, and economic value?
- Do programmes weight signals differently when they define value differently?

The project separates the **selection mechanism** from the grant itself.

Theoretical framework: selection signals as policy instruments

The project treats selection signals as **proxies for expected public value**.

Policy objective $\rightarrow$ Selection signals $\rightarrow$ Funding decision $\rightarrow$ Outcome layers

- Programmes define value differently: innovation, spillovers, or local economic value.
- Granting authorities translate these objectives into signal weights.
- The empirical question is whether these weights predict the outcomes they are meant to generate.

The grant is therefore not the only treatment; the selection rule is part of the mechanism.

Empirical framework

Signals → Weights → Decision → Outcomes

The unit of analysis is the **firm–application**. The project links three layers of data:

- **Selection signals:** founder, project, technology, and regional characteristics
- **Selection rule:** how programmes weight these signals in the funding decision
- **Outcome layers:** innovation, spillover potential, and economic value

The empirical object is not the grant alone, but the link between **what programmes select for** and **what selected firms later generate**.

Identification strategy

Identification uses **variation in selection rules**.

- **Main:** selection model + panel outcome analysis
- **Cleanest causal design:** RDD at rankings or score cutoffs
- **Extensions:** Diff-in-Diff / event study where timing varies
- **Fallback:** IV only with credible external variation

Goal: estimate the **selection–outcome alignment gap**.

Expected contribution

The project contributes to the economics of **public innovation policy design**.

- **Empirical:** evidence on which selection signals predict public value
- **Conceptual:** separates the selection rule from the grant itself
- **Policy-relevant:** identifies gaps between programme objectives and outcomes

Backup: Data Sources

- Programme documents, eligibility criteria, scoring rubrics and beneficiary lists.
- Applicant-level award data from CORDA / ORBIS where accessible.
- Patent data from EPO / PATSTAT.
- Firm accounts, business registers and regional employment data.
- Score-level data from programmes such as the EIC Accelerator where available.

Backup: Identification Risks

- Self-selection into application.
- Score manipulation near cut-offs.
- Missing or incomplete score-level data.
- Survivorship bias in firm-level outcomes.
- Outcome correlation across innovation, spillover and economic-value layers.